

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0864

Parkin 138 kV
October 03, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0864. The Interconnection Customer proposes a 129.9 MW solar photovoltaic facility interconnecting at the Parkin 138 kV substation in the Virginia area. The request was received on September 15, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 385 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,365,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
NEW LONDON to BARKSDALE 138 kV L	WYTHE to S CREEK 138 kV	112.5	245	5.3
PERSON to RIVERVIEW SUBSTATION 345 kV	LEESVILLE to CLOVER 345 kV	111.9	1,286	16.3
WYTHE to UNITY 138 kV Line	WYTHE to SOUTH CRK 138 kV	111.2	288	44.5
BLUFF CITY to LAKE LURE 138 kV	FLUVANNA P.S to LOUISA GENERATION FACILITY 345 kV	103.1	595	8.7
BOONSBORO to NACOOCHEE 138 kV	PARKIN to OPOSSUM CRK 138 kV	103.0	555	51.7
S TEXA to NAOOCHEE SUBSTATION 138 kV line	PARIN to OPOSSUM CRK 138 kV	102.8	557	51.7
BOONSBORO to AMARGO 138 kV	PARKIN to OPOSSUM CRK 138 kV	101.5	857	100.0
UNITY to FLOWOOD 138 kV	SOUTH CRK SUBSTATION to WYTHE 138 kV	101.2	296	44.5
BROOKVILLE to AMARGO 138 kV line	OPOSSUM CRK SUBSTATION to PARKIN 138 kV	101.2	881	100.0
SOUTH LYNCHBURG to BROOKVILLE 138 kV Line	PARKIN SUBSTATION to OPOSSUM CRK 138 kV Line	100.5	920	100.0
BEAR SUBSTATION to CLOVER 345 kV	PERSON to MECKLENBURG POWER STATION 345 kV	100.5	2,871	25.5
SOUTH LYNCHBURG to CNDLERS MTN 138 kV line	PARKIN to OPOSSUM CREEK 138 kV	100.4	931	100.0
CENTRAL TIE to WYTHE 138 kV	PARKIN to CANDLERS MOUNTAIN 138 kV	100.4	932	100.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BLUFF CITY to REACH 138 kV	RED HILL 345 kV/138 kV Auto	100.2	344	5.9
CENTRAL TIE to OPOSSUM CREEK 138 kV	PRKIN to CANDLERS MTN 138 kV	100.2	944	100.0
PARKIN to CANDLERS MOUNTAIN 138 kV	PARKIN SUBSTATION to OPOSSUM CRK 138 kV	100.0	955	100.0
PARKIN to OPOSSUM CREEK 138 kV	PARKIN to CANDLERS MTN 138 kV	100.0	955	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NEW LODON to BARKSALE 138 kV	Rebuild line with higher-capacity conductor	\$6,979,000	8% (\$564,000)
PERSON to RIVERVIEW SUBSTATION 345 kV	Rebuild line with higher-capacity conductor	\$106,039,000	2% (\$2,383,000)
WYTHE to UNITY 138 kV	Rebuild line with higher-capacity conductor	\$12,762,000	12% (\$1,585,000)
BLUFF CITY to LAKE LURE 138 kV	Reconductor line	\$9,506,000	3% (\$259,000)
BOONSBORO to NACOOCHEE 138 kV	Reconductor line	\$2,209,000	11% (\$250,000)
S TEXA to NACOOCHEE SUBSTATION 138 kV	Reconductor line	\$961,000	11% (\$109,000)
BOONSBORO to AMARGO 138 kV	Reconductor line	\$1,085,000	14% (\$147,000)
FLOWOOD to UNITY 138 kV L	Reconductor line	\$2,633,000	12% (\$327,000)
BROOKVILLE to AMARGO 138 kV line	Reconductor line	\$241,000	14% (\$33,000)
SOUTH LYNCHBURG to BROOKVILLE 138 kV Line	Reconductor line	\$694,000	14% (\$94,000)
BEAR SUBSTATION to CLOVER 345 kV	Reconductor line	\$31,114,000	2% (\$759,000)
SOUTH LYNCHBURG to CANDLERS MTN 138 kV Line	Reconductor line	\$1,210,000	14% (\$164,000)
CENTRAL TIE to WYTHE 138 kV	Reconductor line	\$744,000	14% (\$101,000)
BLUFF CITY to REACH 138 kV	Reconductor line	\$10,844,000	3% (\$278,000)
CENTRAL TIE to OPOSSUM CRK 138 kV	Reconductor line	\$988,000	14% (\$134,000)
PARKIN to CANDLERS MOUNTAIN 138 kV line	Reconductor line	\$638,000	14% (\$87,000)
PARKIN to OPOSSUM CREEK 138 kV	Reconductor line	\$670,000	14% (\$91,000)
	Total allocated to project		\$7,365,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.